

Ming Zhou

Tech Lead · Backend & Distributed Platforms · AI Agent Systems

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SUMMARY

Backend and platform engineer with 12+ years across financial services and large-scale internet (Citi, eBay). For the past six years I have been Tech Lead at Citi, where I built and operated a production AI Agent platform on LangGraph, LlamaIndex, and Milvus. The platform serves regulated workflows across multiple business lines in a multi-region environment with hard requirements around reliability, traceability, and data isolation.

I work hands-on across architecture, code, and operations. Strong fundamentals in distributed systems, API design, SLO/SLI, observability, and safe delivery via canary releases and rollback. I partner closely with Product to decide what to build, mentor senior engineers, and operate comfortably across global, distributed teams. Bilingual in English and Mandarin.

KEY PROJECTS (CITI, 2019 – 2025)

AI Agent Platform — Governable RAG & Agentic Workflows

LangGraph orchestration · Auditable RAG · Tool calling · Structured outputs · Human-in-the-loop · SRE-grade reliability

Led the design, delivery, and production operation of an end-to-end AI Agent platform. Turned expert-dependent, manual workflows into a self-service, replayable, risk-controlled human–AI system suitable for regulated, multi-region use.

- Built the platform from 0 → 1. Standardized orchestration, retrieval, policy gating, and audit logging as shared infrastructure. Scaled to 3–8 business lines and 6–12 domains; onboarded 20K–50K governed artifacts and 100–300+ templates.
- Made citation and structured outputs a platform contract (“no evidence, no conclusion”). Held citation coverage at 95–99%; reduced downstream rejections for non-verifiable references from ~12–18% to ~6–10%.
- Designed deterministic safety controls for LLM uncertainty: answerability + risk routing (answer / refuse / citations-only / escalate). Held escalation at 5–15%; reduced unsupported or mismatched answers from ~2–4% to ~1–2%.
- Made version, jurisdiction, and entitlements first-class constraints in retrieval and citation. Wrong-version or unauthorized references dropped from 15–25/month to 3–8/month.
- Productionized to SRE standards: SLOs, distributed tracing, audit logs, canary + rollback, runbooks. P1/P2 incidents 6–10 → 3–6/month; MTTR 60–90m → 25–45m; downtime impact down ~40%.
- Established governed model lifecycle: replay-based evaluation, regression gates, promotion/rollback. Lightweight alignment (7B QLoRA on de-identified data) cut structured-format errors from ~12–20% to ~4–8%.

Stack: Python, FastAPI, LangGraph, LlamaIndex + Milvus, Redis, OpenTelemetry, Jaeger, Langfuse, vLLM, LMDeploy, Prometheus / Grafana, Kubernetes, Helm, Istio, XTuner (QLoRA).

Workflow Automation Agent — Plan-and-Execute Across Internal Systems

Tool calling · Least-privilege retrieval · Structured evidence packages · End-to-end traceability

Owned the delivery of an agent that automates multi-step workflows across 10–20 internal systems and APIs. Standardized a controllable, replayable, audit-ready pipeline that cut cycle time, rework, and cross-system data exposure.

- Built a reusable plan-and-execute workflow engine. Multi-step tasks became a governed, replayable pipeline under least-privilege constraints.
- Designed the connector layer with execution controls (allowlists, step budgets, retries / timeouts, circuit breakers, tiered degradations). Avg steps 18–25 → 10–16; p95 runtime 25–40m → 12–25m.
- Integrated 10–20 internal systems and APIs (tickets, approvals, access/change logs, reports, alerts, config snapshots). Reduced “wrong period / version / not found” rework from 40–70/month to 20–45/month.
- Shifted quality left with validation gates (time window, owner, approval chain, authoritative source, conflict detection). Downstream rejections 18–28% → 10–18%.
- Standardized output package schema with minimum-disclosure principles (IDs, fields, timestamps, hashes). Manual redaction load 10–16% → 5–10%.
- Scaled to 200–500 active users and 3K–10K runs/month. Time-to-root-cause 2–4h → 0.5–1.5h via full traceability.

Stack: Python, FastAPI, LangGraph, internal HTTP/gRPC APIs, OpenTelemetry, Jaeger, Langfuse, Prometheus / Grafana, Kubernetes, Helm, Istio.

Decision Support Agent — Maker-Checker & Human-in-the-loop

Role-based collaboration (Planner / Executor / Critic) · Maker-Checker · Decision arbitration · Auditable replay

Built a decision support agent for high-risk decision points. Converted experience-driven, scattered approvals into a standardized maker–checker workflow with clear human accountability.

- Built a maker–checker decision workflow framework with “no sign-off, no issuance” and auditable trails.
- Codified executable review criteria and arbitration rules (approve / rework / escalate) for missing or conflicting evidence and entitlement / jurisdiction blocks. Median re-review cycles 2 → 1.
- Raised high-risk human sign-off coverage from ~70–80% to ~95–99% with automated second-line checks and tiered approvals.
- Standardized structured review packages. Cross-team dispute / variance dropped from 15–25% to 5–10%.
- Cycle time to review-ready outputs: p50 2–4h → 45–90m; p95 1–2 business days → 4–8h.
- Operationalized replay and governance: trace_id, role chain, rule hits, evidence IDs, arbitration outcomes, approvals — fed back into replay / regression suites.

Stack: Python, FastAPI, LangGraph, OpenTelemetry, Jaeger, Langfuse, Prometheus / Grafana, Kubernetes, Helm, Istio.

WORK EXPERIENCE

Citi, Shanghai — Assistant Vice President / Tech Lead Jul 2019 – Sep 2025

- Led architecture and delivery for an AI Agent / RAG platform serving regulated, audit-sensitive workflows. Combined LangGraph / LlamaIndex / Milvus with deterministic gating, explainability traces, and controlled escalation paths.
- Owned platform reliability and release governance: SLO/SLI, alerting and runbooks, canary releases via Istio, CI/CD hardening, and security scanning.
- Set engineering standards across RFCs, design reviews, and post-incident learning. Mentored senior engineers and partnered with Product on solution shape.

eBay (AWF), Shanghai — Senior Full-Stack Engineer Oct 2018 – Jul 2019

- Built and ran a real-time log analytics and alerting platform for high-volume telemetry. Led the MySQL → TiDB migration and Kubernetes-based deployment for scalability.
- Strengthened production observability (Prometheus / Grafana) and incident-response workflows in a large-scale e-commerce environment.

NiWoDai, Shanghai — Senior Full-Stack Engineer Jul 2018 – Oct 2018

- Delivered backend services for lending workflow systems. Focus on throughput / latency tuning and production hardening (timeouts/retries, idempotency, operational readiness).

Citi, Shanghai — Senior Full-Stack Engineer Aug 2016 – Jul 2018

- Designed and delivered distributed services for internal finance and operations platforms on Spring Boot / Spring Cloud. Focus on API architecture, data integration, and reliability foundations (monitoring, runbooks, on-call).

Earlier Roles — Full-Stack / Senior Engineer Dec 2012 – Jun 2016

ODianYun, Wistron, Beijing Murong Technology. Built enterprise applications and backend services; established fundamentals in system design, data stores, deployments, and production support.

EDUCATION

East China Jiaotong University of Science & Technology B.S., Computer Science and Technology 2009 – 2013

TECHNICAL SKILLS

Languages: Java, Python. Familiar with TypeScript / Node.js.

LLM / AI Engineering: LangGraph, LangChain, LlamaIndex, Milvus, hybrid retrieval (vector + BM25), reranking, metadata versioning, tool / function calling, structured outputs, evaluation and replay, guardrails (PII minimization, prompt-injection defenses), OpenAI API, Hugging Face

(Transformers, LoRA / QLoRA), InternLM XTuner, vLLM, LMDeploy (Turbomind / PyTorch / TensorRT-LLM).

Backend & APIs: Spring Boot, Spring Cloud, FastAPI, Flask, Gunicorn, RESTful microservices, API design and integration, event-driven architecture.

Distributed Systems & Reliability: high availability, low latency, SLO / SLI, alerting and runbooks, incident debugging, canary releases, traffic management, timeouts / retries / circuit breakers, idempotency.

Databases & Caching: MySQL, PostgreSQL, TiDB, MongoDB, Redis.

Cloud Native & DevOps: Docker, Kubernetes, Helm, Istio, Jenkins, GitHub Actions, hardened images, CI/CD with automated testing and security scanning.

Observability: Prometheus, Grafana, OpenTelemetry (Collector), Jaeger, Langfuse.

Developer Tooling: Claude Code, Cursor, GitHub Copilot, n8n. Daily use across coding, code review, testing, documentation, and engineering productivity.